

# **CariSurg MedTech Pathways Research Project**

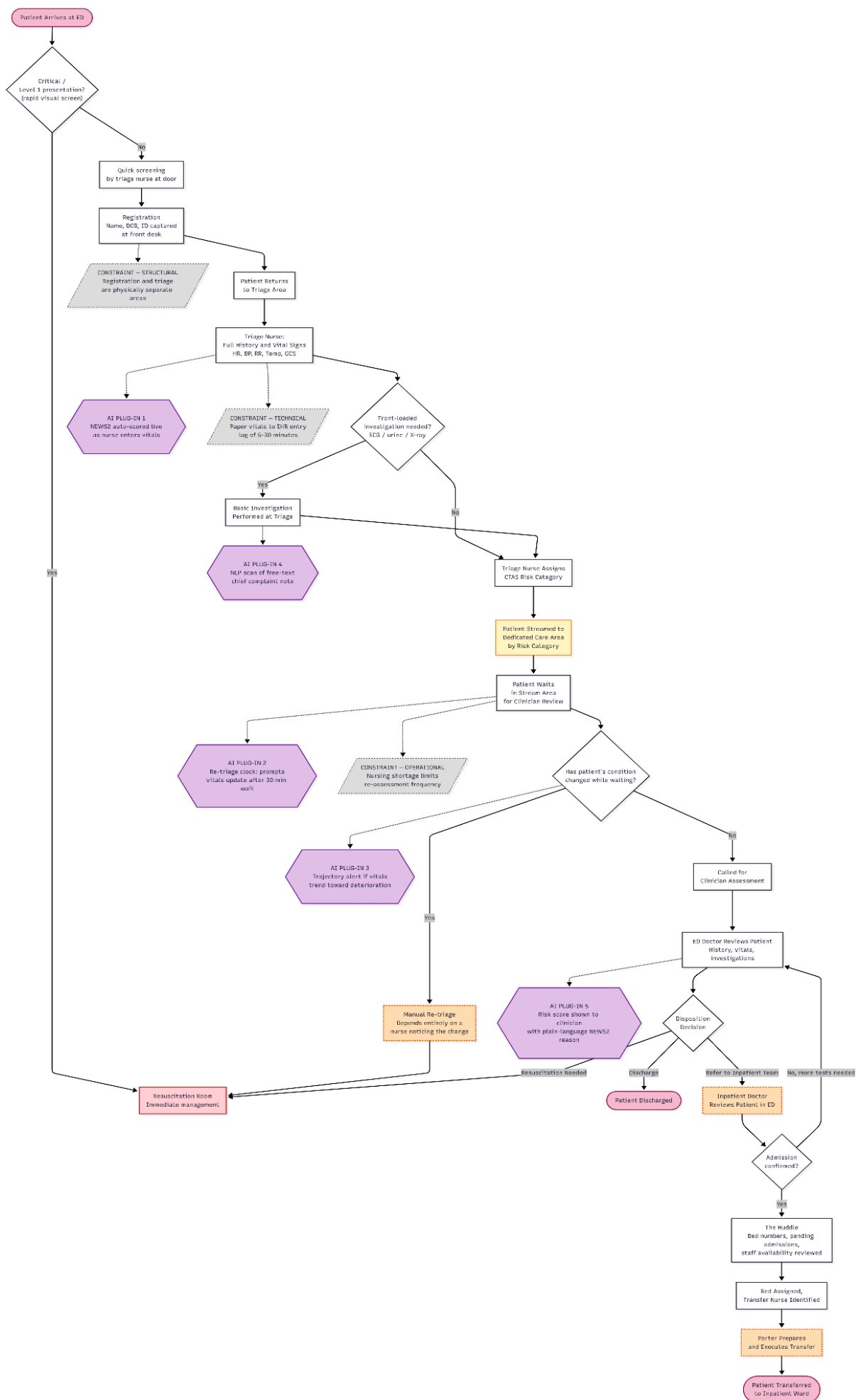
**Project Title:** Systems Thinking: Workflow Mapping, AI Plug-In Points, and Constraints

**Student Name:** Ashi Baldie

## **Methodology**

The workflow diagram below was adapted directly from the validated process map structure published by <sup>[3]</sup>, the only qualitative observational study of patient flow conducted in a Caribbean ED to date. That study mapped 143 patient journeys across 203 hours of direct observation at a public teaching hospital ED in Trinidad and Tobago, and identified six categories of organisational factors that facilitate or hinder flow: work processes, design and layout, material resources, nursing staffing, non-clinical staff, and external departments <sup>[3]</sup>. Building the Mercer General workflow on this validated structure, rather than a textbook ED flowchart, ensures the diagram reflects how Caribbean EDs actually operate rather than how Western literature assumes they operate.

## **Workflow diagram**



**Figure 1.** Show the flow diagram tracing the patient journey from arrival to ward transfer using the same structural backbone as <sup>[3]</sup>'s main process map (streaming and triage, front-loaded investigations, the transfer "huddle"), with five AI plug-in points overlaid in purple and three workflow constraints overlaid in grey.

See triage workflow diagram in plots/ on

<https://github.com/baldiea75-tech/carisurg-portfolio.git>

## The AI plug-in

**AI Plug-in 1: Live NEWS2 scoring at triage.** As the triage nurse enters vitals, the system calculates a NEWS2 score in real time rather than requiring a separate manual calculation step.

**AI Plug-in 2: Automated re-triage clock.** If a patient has waited more than 30 minutes in the stream area without a vitals update, the system issues a visual prompt. This directly targets the static snapshot problem: AI decision support tools that update only once at admission show strong predictive performance on retrospective data but are evaluated using a single point-in-time input, leaving deterioration during the wait undetected <sup>[8]</sup>.

**AI Plug-in 3: Trajectory alert.** Rather than scoring a single reading in isolation, the system flags a worsening trend across consecutive readings, similar in principle to the 10-minute interval re-evaluation approach validated by <sup>[8]</sup>, who found that updating predictions throughout a patient's ED stay rather than once at arrival substantially improved the clinical relevance of an admission-prediction model across 154,347 ED visits.

**AI Plug-in 4: NLP scan of free-text complaints.** A lightweight natural language scan of the chief complaint note flags high-risk language. This plug-in is justified by <sup>[1]</sup>, who demonstrated that integrating structured vitals with unstructured free-text triage notes improved hospitalisation prediction over structured data alone.

**AI Plug-in 5: Plain-language risk explanation at clinician review.** Rather than displaying a raw score, the system maps the output onto NEWS2 language clinicians already use. This design choice is grounded in evidence that clinician trust in AI output depends heavily on how the recommendation is communicated, not only on model accuracy <sup>[5]</sup>.

## Workflow Constraints

**Constraint 1: Physical separation of registration and triage.** <sup>[3]</sup> found that the physical separation of registration and triage areas created additional waiting and activity steps that did not exist in the clinical process itself, purely as a consequence of department layout. Any AI system assuming a single continuous data-capture point at "arrival" must account for this gap, since vitals are not captured until the patient physically reaches the second area.

**Constraint 2: Nursing staff shortages limiting re-assessment capacity.** <sup>[3]</sup> documented that EDs in Caribbean hospitals frequently operate below the nurse staffing levels required for the streaming model to function as designed, forcing nurses to multitask across multiple care streams simultaneously.<sup>[4]</sup> similarly found in a systematic review of AI's influence on clinical work design that AI tools succeed only when they reduce rather than add to existing workload, since understaffed clinical environments have no capacity absorb additional manual steps. <sup>[9]</sup> reinforced this through 200 hours of observation and 21 nurse interviews in a tertiary hospital, finding that the most consistently cited barrier to nursing CDSS adoption was poor fit between the system and existing task demands, not the underlying algorithm. Any re-triage or alerting system must therefore generate output passively rather than requiring active nurse interaction to function.

**Constraint 3: Competing demands on porter and transfer staff.** <sup>[3]</sup> identified that porters were frequently pulled between transferring admitted patients to wards and transporting patients to investigations, creating conflict over which task should take priority and consequently delaying both. This is a downstream constraint rather than a triage-stage constraint, but it illustrates that flow-level AI interventions at the front of the department cannot resolve outflow bottlenecks further down the pathway; the digital triage tool addresses input and throughput but does not eliminate output-stage delays, a limitation consistent with the three-component ED crowding model used by <sup>[8]</sup>, who note that input, throughput, and output factors interact and that solving one does not automatically resolve the others.

## **Additional implementation evidence**

Beyond the three named constraints, two further studies strengthen the case for the specific design choices proposed in this project. <sup>[6]</sup> conducted a human-factors workflow integration analysis of a clinical decision support tool deployed directly in an emergency department, drawing on 134 barrier and facilitator excerpts from 12 ED physicians, and found that adoption depended heavily on whether the tool fit into the existing timing, flow, and patient journey scope of clinical work rather than requiring a separate interaction step, directly supporting the design requirement in Constraint 2 that the system surface passively rather than add interaction steps. <sup>[7]</sup> extended this finding in a post-implementation usability evaluation of the same tool, identifying that despite high perceived usability in simulation, real-world clinical use remained low, with poor workflow integration cited as the dominant barrier, reinforcing the rationale for AI Plug-in 5's plain-language, NEWS2-aligned explanation surfaced directly within the existing clinician review step rather than as a separate system.

## Stakeholders

| Stakeholder                 | Top concern                                                                                                                                                                               |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Triage nurse                | The system must not add steps to an already understaffed, interruption-heavy workflow                                                                                                     |
| ED consultant               | Alerts must be clinically actionable, not noise; alert fatigue research shows that systems generating high false-positive rates are quickly ignored or overridden by staff <sup>[4]</sup> |
| Registration clerk          | Free-text chief complaint quality directly affects NLP plug-in accuracy                                                                                                                   |
| Clinical IT Lead            | Full audit trail, data governance compliance, zero real patient data in pilot                                                                                                             |
| Patient and family advocate | Assurance that automated scoring does not delay care for patients who present calmly but deteriorate quietly                                                                              |

## *References*

1. Arnaud E, Elbattah M, Gignon M, Dequen G. Deep Learning to Predict Hospitalization at Triage: Integration of Structured Data and Unstructured Text. In: 2020 IEEE International Conference on Big Data (Big Data). IEEE; 2020:4836-4841.  
doi:10.1109/BigData50022.2020.9378073
2. Boonstra A, Laven M. Influence of artificial intelligence on the work design of emergency department clinicians a systematic literature review. BMC Health Serv Res. 2022;22(1):669. doi:10.1186/s12913-022-08070-7
3. De Freitas L, Goodacre S, O'Hara R, Thokala P, Hariharan S. Qualitative exploration of patient flow in a Caribbean emergency department. BMJ Open. 2020;10(12):e041422.  
doi:10.1136/bmjopen-2020-041422
4. Olakotan OO, Mohd Yusof M. The appropriateness of clinical decision support systems alerts in supporting clinical workflows: A systematic review. Health Informatics J. 2021;27(2):14604582211007536. doi:10.1177/14604582211007536
5. Perivolaris A, Adams-McGavin C, Madan Y, et al. Quality of interaction between clinicians and artificial intelligence systems. A systematic review. Future Healthcare Journal. 2024;11(3):100172. doi:10.1016/j.fhj.2024.100172
6. Salwei ME, Carayon P, Hoonakker PLT, et al. Workflow integration analysis of a human factors-based clinical decision support in the emergency department. Applied Ergonomics. 2021;97:103498. doi:10.1016/j.apergo.2021.103498
7. Salwei ME, Hoonakker P, Carayon P, Wiegmann D, Pulia M, Patterson BW. Usability of a Human Factors-based Clinical Decision Support in the Emergency Department: Lessons Learned for Design and Implementation. Hum Factors. 2024;66(3):647-657.  
doi:10.1177/00187208221078625
8. Van Der Haas Y, Roskamp W, Chang-Willems LEM, et al. Evaluating an AI Decision Support System for the Emergency Department: Retrospective Study. JMIR AI. 2026;5:e80448. doi:10.2196/80448
9. Zhai Y, Yu Z, Zhang Q, Zhang Y. Barriers and facilitators to implementing a nursing clinical decision support system in a tertiary hospital setting: A qualitative study using the FITT framework. International Journal of Medical Informatics. 2022;166:104841.  
doi:10.1016/j.ijmedinf.2022.104841